Lecture plan · for the lecturer

Integer programming and branch and bound

Optimization · University · Lecture 2

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Model with integers.
- Use relaxations as bounds.
- Run branch and bound.

Activity: Grow the tree

1. Warm-up (5 min): Show 2.5 buses and ask students to vote: round up, round down, or something else.
2. Relax (10 min): Solve the knapsack relaxation by value-to-weight ratio.
3. Branch (25 min): Pairs draw the tree, computing bounds and pruning nodes.
4. Compare (15 min): Compare the rounded answer, the proven optimum and the gap.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How many nodes did you prune?
- How far was rounding from optimal?
- What does the gap tell a manager?

Exit ticket

- Name one decision in your institution that needs integer variables.

Try it in your next lecture

Give pairs a four-item knapsack and ask them to round the relaxation first; then draw the branch-and-bound tree together and compare the rounded answer with the proven optimum.

Seminar sheet · for students

Integer programming and branch and bound

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Name _____ Class _____ Date _____

1 **Best whole solution 13; a branch's relaxation gives 12. What next?**

◯ Branch further ◯ Prune it ◯ Round it

2 **Solve the knapsack.**

Items (weight, value): a (5, 10), b (4, 7), c (3, 6); capacity 7. The LP relaxation gives _____ and the integer optimum is _____.

Word bank: 14 · 10 · 16 · 13

3 **Which statements about branch and bound are true? Choose all that apply.**

☐ It branches on a fractional variable ☐ It prunes nodes whose bound cannot beat the incumbent

☐ It always enumerates every combination ☐ The bound–incumbent gap certifies quality

4 **Rounding down an optimal LP relaxation always gives an optimal integer solution.**

◯ True ◯ False

5 **Order one branch-and-bound iteration.**

☐ Update the incumbent if integral

☐ Prune, or branch on a fractional variable

☐ Compare its bound with the incumbent

☐ Solve the node's relaxation

✎ **Do your students ever see a rounded solution fail? Which example from your context would show it?**